GRIMIT LABORATORY FURNITURES AND EQUIPMENT CORP.

PROFORMA QUOTATION

PARTIDO STATE UNIVERSITY Camarines Sur			**Ref. No. GLFEC – 2022-006-01-001**	
Attn.: **ENGR. KIER P. GASGA** **Research and Development Coordinator**			Date: January 24, 2022	
No.	Qty.	Description	Unit Price	Total Amount
1	1 unit	**UV-VIS Spectrophotometer** SHIMADZU UV-1900 Country of Origin : Japan Wavelength range: 190 to 1100 nm Spectral bandwidth: 1nm (190 to 1100 nm) Wavelength display: 0.1-nm increments Wavelength setting: 0.1-nm increments (1-nm increments when setting scanning range) Wavelength accuracy: ±0.1 nm at 656.1nm D2 ±0.3 nm for entire range Wavelength repeatability: ±0.1 nm Wavelength slew rate : About 14,500 nm/min Wavelength scanning speed: 3,000 to 2 nm/min 29,000 nm/min when survey scanning Lamp Interchange Wavelength: Automatic interchange linked to wavelength. The interchange can be set freely in the range of 295 to 364 nm (0.1 nm increments) Stray light: less than 0.02% NaI at 220 nm, less than 0.02% NaI at 340 nm (NaNO2) less than 0.5% at 198 nm (KCI) Photometric system: Double Beam optics Photometric range: Absorbance: -4 to 4 Abs Transmittance: 0% to 400% Photometric accuarcy ±0.002 Abs at (0.5 Abs) ±0.004 Abs at 1.0 Abs ±0.006 Abs at 2.0 Abs Photometric repeatability: less than ±0.002 Abs at 0.5 Abs less than ±0.002 Abs at 1 Abs less than ±0.001 Abs at 2.0 Abs Baseline stability: less than 0.0003 Abs/H at 700 nm (one hour after light source turned ON) Baseline flatness: within ±0.0006 Abs (190 to 1100 nm, one hour after light source turned ON) Noise level: less than 0.00005 Abs (700 nm) Light source: 20-W halogen lamp and deuterium lamp Built-in light source auto position adjustment Monochromator: Blazed holographic grating in Czerny-Turner mounting Detector: Silicon photodiode Sample compartment: Internal dimensions: W110 × D250 × H115 mm Distance between light beams: 100 mm Power requirements: AC100,120,220,230,240 V, 50/60 Hz, 140 VA Environmental requirements: Temperature: 15°C to 35°C Humidity: 30% to 80% (without condensation; 70% max. at 30°C or higher) Dimensions: W450 × D490 × H270 mm Weight: 15 kg Output device: USB memory (optional) Data files saved in text format or UVPC format. UVPC-format files can be read directly	**P 800,000.00**	**P 800,000.00**

Eastpoint Business Center No. 4 East Service Road AFPOVAI Western Bicutan Taguig City 1633 Metro Manila Philippines
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www.grimitlab.com

		by UVProbe and LabSolutionsTM UV –VIS PC compatibility: UVProbe software. LabSolutionsTM UV –VIS Software External control possible via USB.		
2	1 unit	**SpectraMax ABS Plus UV/Vis Single-Mode Microplate Reader with built-in Cuvette Port** **System includes:** Tunable Monocromator-based Spectral Absorbance (190-1000nm) SoftMax Pro Software (4 perpetual licenses) Integrated Linear Shaking and Temperature Control from ambient +50C to 450C Power Cable and USB Cable Other Specifications: Compatible with 96- and 384-well plates Compatible with SpectraDrop Micro-Volume Microplate Warranty: 12 months on parts warranty 12 months MDCare PhD Technical Support Plan	**P 1,976,541.00**	**P 1,976,541.00**
3	1 unit	**SCI100-Pro Digital Rotary Evaporator** Features: Digital display of both rotation speed and heating temperature allows for optimal control of all distillation processes 5L heating bath with wide temperature range from room temp. to 180°C Patented condenser (cooling surface 1500cm²) with excellent cooling effect Automatic motor lift releases the evaporating flask to a safe position in case of power failure Speed range from 20 to 280rpm, and interval operation in clockwise and anticlockwise directions for drying process Overheating protection temperature at 220°C Timer function for precise control processing Chemical-resistant PTFE seals and patented pressure springs provide excellent sealing Precise temperature control and adjustable safety circuit Quick release connectors for easy flask removal Ejection mechanism ensures easy removal of evaporating flask Patented double spring sealing ring which is made of PTFE provides excellent sealing performance Remote function provides PC control and data transmission Specification: Motor Type: Brushless DC Motor Speed Range: 20-280rpm Display LCD (speed, temperature, time) Clockwise and Anti-clockwise Yes Heating Temperature Range: Room temp. To 180°C Control Accuracy: ± 1°C Heating Power: 1300W Stroke Displacement: automatic 150mm Timer: Yes Time Setting Range: 1-999min Dimension [DxWxH]: 465x467x583mm Weight: 15kg Permissible Ambient Temperature: 5-40°C Permissible Relative Humidity: 80%RH Protection ClassIP20 USB Interface: Yes	**P 400,000.00**	**P 400,000.00**

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		Voltage/Frequency: 110V, 60 Hz Power 1400 W		
4	1 unit	**UW/UX Series Electronic Balances** Shimadzu Model UX420H Capacity (g) 420 Minimum Display (g) 0.001 Calibration Range with External Weights (g) : 100–420 Repeatability (σ) (g) ≤0.001 Linearity (g): ±0.002 Response Time (s) 1.5–2.5 Ambient Temperature and Humidity: 5–40°C 30–85% (No Condensation) Temperature Coefficient Sensitivity (ppm/°C) (10–30°C) : ±3 Pan Size (mm) approx. 108 × 105 Main Body Dimensions (mm) approx.: W190 × D317 × H78 Weight (kg) approx.: 2.7 Display : LCD with backlight Power Requirements: DC, 10 to 15.5 V, 500 mA Data I/O: RS-232C Features LabSolutions Balance WindowsDirect GLP/GMP/ISO conformance Analog display, % display, PCS, User unit, Animal weighing, Speci¬c gravity measurement S/W, Checkweighing	**P 80,000.00**	**P 80,000.00**
5	1 unit	**Digital Compact Hotplate Stirrer** Daihan Scientific Model: MSH-30D Patented Jog-Shuttle Control System • Speed & Control Resolution: 80 ~ 1,500 Rpm, 5 Rpm • Temp. Accuracy / Resolution: ±0.3°C At Set Temp. / 0.1°C-Display, 0.5°C-Control • Timer & Alarm: 99hr 59min(With Continuous Run), Error Status & Timer-End • Safety Circuit: Over Temp. Protection • User's Self-Compensation: Control Difference Between Real Temp. & Indicated Temp. (±20.0°C) • Accurate Temp. Control of The Sample In The Vessel Using PT100 Sensor • Storage Function for the set Values: Temp. and Timer • Locking Mode Supported for Experimental Safety (Input to Jog-Shuttle can be Disabled)	**P 70,000.00**	**P 70,000.00**
6	1 unit	**Centrifuge 5430, keypad, non-refrigerated** without rotor, 230V/50-60Hz (EU) Include: Rotor for F-35-6-30, fixed-angle including rotor lid, with 6 adapters each for 15/50mL conical tubes	**P 833,806.00**	**P 833,806.00**
7	1 unit	**Centrifuge 5920R, keypad, refrigerated, with** Rotor S-4 x 1000 Universal-Large including universal buckets and adapters for round-bottom tubes with diameter 13/16mm, 230V/50-60Hz (EU) Additional Adapters for Rotor S-4 x 1000 Universal-Large: Adapters for Rotor, for 49 tubes, 2.6-8mL, set of 2pcs. 2 Adapters for Rotor, for 37 tubes, 5.5-12mL, set of 2pcs. Adapters for Rotor, for conical tubes, 5/15mL 2 set of 2pcs.	**P 3,276,329.00**	**P 3,276,329.00**

		Adapters for Rotor, for conical tubes, 25/50mL 2 and plates, set of 2pcs. Plate carrier, set of 2 pcs. Additional Rotor: Rotor S-4 x 1000 Round buckets, aerosol-tight 1 Adapters for 1 bottle 1L Nalgene®, set of 2pcs.		
8	1 unit	**Shaker Incubator** **BIOBASE Model BJPX-1102D** Controller: PID Display: LCD Circulation Mode: Forced convection Shaking Mode: Rotary shaking Drive Mode: Multidimensional drive Rotating Speed: 20-350 rpm (Static culture, reverse rotation or alternate operation) Rotating Accuracy: ±1rpm Vibration Range: Φ26mm, (Can customize φ35(20-250rpm/min), φ50(20-200rpm/min)) Temp. Range: 5-65°C Temp. Accuracy: ±0.1°C Temp. Uniformity: ±0.5°C(At 37°C) Timing Range 0-9999h/min Shaking Plate Size: 500*360mm Shaking Plate Quantity: 2pcs Max. Capacity: 100ml*60 or 250ml*38 or 500ml*22 or 1000ml*16(Flask clamp) 100ml*60 or 250ml*36 or 500ml*24 or 1000ml*16 (Haftplatte) Standard Capacity: 250ml*24 Internal Material: Stainless steel External Material: Stainless steel Cooling Consumption: 250W Heating Consumption: 500W	**P 150,000.00**	**P 150,000.00**
9	1 unit	**Drying Oven** Biobase Model BOV-T30C Capacity: 43L Temp. Range: 50-200°C Temp. Fluctuation: ±1°C Ambient Temp.: 5-40°C Timing Range: 1-9999min Shelves No.: 2-3 pcs Power Consumption: 850W Power Supply: Standard 220V±10%, 50/60Hz	**P 150,000.00**	**P 150,000.00**
10	1 unit	**Ducted Fume Hood FH(P)** **BIOBASE Model FH1000(P)** **Advantage:** 1. It is safer to use anti-corrosive water tap. 2. Microprocessor control system, LED display 3. With memory function in case of power-failure 4. Made of porcelain white PP, resistant to acid, alkali and anti-corrosion. 5. Front window which is made of thick transparent toughened glass maximize light and visibility inside the fume hood, providing a bright and open working environment. **Specification:** External Size (W*D*H): 1047*800*2450mm	**P 400,000.00**	**P 400,000.00**

		Internal Size (W*D*H): 787*560*700mm Work Surface Height: 820mm Max Opening: 740mm Air Velocity: 0.3~0.8m/s Noise: ≤68dB Exhaust Duct: PVC, Standard length: 4 meters. Φ300mm LED Lamp: 24W*1 Blower Built-in PP centrifugal blower(2 blowers for FH1800(P) only); Speed adjustable Front Window: Resistant to acid and alkali, Manual, 5mm toughened glass, height adjustable. Power Supply: AC220V±10%, 50/60Hz; 110V±10%, 60Hz Consumption: 400W Materia Main Body Made of Porcelain white PP, thickness 8mm, resistant to strong acid, alkali and anti-corrosion. Work Table: Chemical resistant phenolic resin Standard Accessory LED light Work table: Chemical resistant phenolic resin; 4.3 meters PVC duct (2 pcs of 4.3 meters PVC dust for FH1800(P) only), Diameter: 250mm; Pipe strap*2(4 pcs for FH1800(P) only); RVV power cord(3*1.0mm²*3000mm); Base cabinet; Waterproof socked*2; Tail insert; Water tap(PP); Gas tap(PP); Water Sink; PP centrifugal blower*1(2 for FH1800(P) only) Optional Accessory PP work table, epoxy resin board or ceramic board; Active carbon filter; Outer PVC centrifugal blower*1(only need to select when the duct more than 4m, 2 for FH1800(P) only); Splicable base cabinet(only not for FH1800(P)) Gross Weight: 137kg Package Size (W*D*H) Main Body: 1188*938*1612mm Base Cabinet: 1188*888*1000mm		
11	1 unit	**AUTOCLAVE** **DAIHAN Model MaXterile 60** -Vertical /Floor Type -Chamber (inner size) Capacity: 60L -Temp Range: ambient to Room Temp +5℃ to 132℃, ±0.5℃ at 121℃ -Operation Pressure: 1.2 kgf/cm2 at 121℃ / 2.0 kgf/cm2 at 132℃ -Air vent: adjustable -Temp Controller: Digital Fuzzy-Control System (Fully Automatic) -Timer & Alarm: 60 min, Error status & Timer-end -Pressure guage: Mechanical type, 0-3kg/cm2	**P 400,000.00**	**P 400,000.00**

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		-Safety Device: over heat Protector, Over Pressure Protector -Monitor Unit: Audible & Visible Device -Basket: Standard Type, 2EA -Power: 220V, 50/60Hz -Inclusions: -2 stainless steel wire basket		
		Total		**P 8,536,676.00**

NOTE: Price firm for 30 days. Prices computed at the present rate of exchange, any increase at the time of order shall be on the account of the buyer.

Returned Products: Products may be returned only after receiving authorization from GRIMITLAB Sales and Marketing Manager. Products returned without authorization may be refused or returned at Client's expense. Authorized returns must be in new and sellable condition in original manufacturer's packaging. The Product must be of current design and finish. Returns will not be allowed for discontinued or obsolete items. Returned Products are subject to a 25% restocking charge unless the merchandise was shipped in error by GRIMITLAB, or if the Products were defective upon receipt from GRIMITLAB Sales and Marketing. Special-ordered, non-inventory Products may not be returned. All returned Products will be inspected by GRIMITLAB Sales and Marketing before credit is issued. Shipping charges for returned Products must be prepaid by You. Products may not be returned after 30 days from the date of invoice. Proof of purchase must be obtained on all items to be returned. Credit will be issued at the actual cost paid as shown on the proof of purchase. The 25% restocking charge will be added from the sales invoice amount.

Place of Delivery: Ilocos Sur
Terms of Delivery: 30 to 60 days
Terms of Payment: 30 days
Price Validity: 30 days
Note: 12% VAT Inclusive

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C101-E154

Typical Specifications

UV-1900

UV-VIS Spectrophotometer

The UV-1900 is a double-beam UV-Vis spectrophotometer using Shimadzu's original LO-RAY-LIGH™ diffraction grating technology. In addition to its high optical performance, the UV-1900 features high resolution, low stray light, high reproducibility, and an ultra-fast scan function. It also has an easy-to-use interface on a color touch-screen display. The UV-1900 is designed to meet the needs of both high performance and usability.



Hardware Specifications

Item	Specification
Wavelength range	190 to 1,100 nm
Spectral bandwidth	1 nm (190 to 1,100 nm)
Wavelength display	0.1 nm increments
Wavelength setting	0.1 nm increments (1 nm increments when setting scanning range)
Wavelength accuracy	± 0.05 nm at D2 peak 656.1 nm, ± 0.3 nm for entire range
Wavelength repeatability	± 0.1 nm
Wavelength slew rate	About 14,500 nm/min
Wavelength scanning speed	3,000 to 2 nm/min 29,000 nm/min when survey scanning
Lamp interchange wavelength	Automatic interchange linked to wavelength. The interchange wavelength can be set freely in the range of 295 to 364 nm (0.1 nm increments).
Stray light	Less than 0.02% at 220 nm (NaI) Less than 0.01% at 340 nm ($NaNO_2$) Less than 0.5% at 198 nm (KCl)
Photometric system	Double beam optics
Photometric range	Absorbance: -4 to 4 Abs Transmittance: 0% to 400%
Photometric accuracy	± 0.002 Abs at 0.5 Abs ± 0.0025 Abs at 1.0 Abs ± 0.006 Abs at 2.0 Abs (measured using NIST930D/NIST1930 or equivalent.)
Photometric repeatability	Less than ± 0.0001 Abs at 0.5 Abs Less than ± 0.0001 Abs at 1 Abs Less than ± 0.0005 Abs at 2 Abs

The specifications shown here represent the average performance of the UV-1900. These specifications are typical values, not guaranteed values. The guaranteed specifications are listed in a separate publication.

Item	Specification
Baseline stability	Less than 0.0003 Abs/Hr (700 nm, one hour after light source turned ON)
Baseline flatness	Less than ± 0.0006 Abs (1,100 to 190 nm, one hour after light source turned ON)
Noise level	Less than 0.00003 Abs (700 nm)
Light source	20-W halogen lamp and deuterium lamp Built-in light source auto position adjustment
Monochromator	LO-RAY-LIGH grade blazed holographic grating in Czerny-Turner mounting
Detector	Silicon photodiode
Sample compartment	Internal dimensions: W110 × D250 × H115 mm Distance between light beams: 100 mm
Power requirements	AC100,120,220,230,240 V, 50/60 Hz, 140 VA
Environmental requirements	Temperature: 15°C to 35°C Humidity: 30% to 80% (without condensation; 70% max. at 30°C or higher)
Dimensions	W450 × D501 × H244 mm
Weight	16.6 kg
Output device	USB memory (optional) Data files saved in text format or UVPC format. UVPC-format files can be read directly by UVProbe and LabSolutions™ UV-Vis.
PC compatibility	UVProbe software (standard) LabSolutions UV-Vis software (optional) External control possible via USB.
Display	24-bit color touch screen
Supported languages	Japanese, English, Chinese, Spanish (Mexico), Portuguese (Brazil).

Software Specifications

Measurement mode	Specification
Photometric mode	Single-wavelength measurement 1. Photometric modes: T% or Abs 2. Quantitation using K-factor method 3. Data table storage and recall functions Multiple-wavelength measurement 4. Photometric modes: T% or ABS 5. Measurements at up to eight designated wavelengths (set in 0.1 nm increments) 6. Data calculation at up to four wavelengths (difference or ratio between two wavelengths, calculation between three wavelengths, etc.) is possible.
Spectrum mode	1. Measurement modes: ABS, T%, E 2. Number of repeat scans: 1 to 99 3. Recording system: Selection between single spectrum and data overlay 4. Data storage and recall 5. Data processing: Peak/valley detection, arithmetic operations, differentiation, smoothing, area calculation, point picking, data reading at cursor-specified point
Quantitation mode	1. Measurement methods: 1-wavelength, 2-wavelength, 3-wavelength, and 1st to 4th derivative methods 2. Quantitation methods: Automatic concentration calculation using K-factor Automatic concentration calculation using single-point calibration curve Multi-point calibration curve method (1st to 3rd order regression curves) 3. Measurement parameters: Number of standards (2 to 10) Number of repeat measurements (1 to 10 times) to obtain a mean value for quantitation.
Kinetics mode	1. Measures absorbance changes as a function of time and calculates the enzymatic activity values. 2. Measurement time: 1 to 9,999 sec/min 3. Measurement methods: 1-wavelength, 2-wavelength, multi-cell, and rate measurements
Time scan mode	1. Measures changes in measured values as a function of time. 2. Measurement mode: ABS, T%, E 3. Measurement time: 1 to 9,999 sec/min 4. Data processing functions (same as spectrum mode)

Measurement mode	Specification
Biomethod mode	DNA/Protein Quantitation 1. Calculation of DNA/protein concentration and absorbance ratio DNA concentration = K1 x A1 - K2 x A2 Protein concentration = K3 x A2 - K4 x A1 2. Factors and measurement wavelengths can be set freely. 3. Background correction is possible. Quantitation of proteins 1. Quantitation methods: Lowry method, BCA method, Biuret method, CBB method (Bradford method), UV method
Maintenance	1. Baseline correction 2. Lamp usage time display and reset. 3. Security settings Functions can be restricted according to the user. 4. Instrument validation functions: 1) Compatible with 9 JIS items Wavelength accuracy, wavelength repeatability, resolution, stray light, photometric accuracy, photometric repeatability, baseline flatness, baseline stability, noise level. 2) Semi-automatic validation Validation inspections conducted interactively while inserting and removing inspection jigs. 3) Fully automatic validation Automatic validation inspections from measurement to evaluation and printout. 4) Setting inspection parameters and pass/fail criteria Authority to make changes can be protected by password access. 5) Detailed printout of results. 6) Bulk printout of results. 7) Equipped with method in accordance with Pharmacopeia (JP, USP, EP).
Shared functions	1. Automatic setting of measurement mode after instrument initialization. It's possible to specify standby and parameter files in the parameter setting window for each measurement mode. 2. Selection of displayed number of decimal places Absorbance: 3 or 4 decimal places Transmittance: 1 or 2 decimal places 3. Number of files that can be saved (built-in memory) Measurement parameters: 100 files max. Tabular data: 15 files max. Curve data: 16 files max. Validation condition: 10 files max. Validation result: 3 files max. 4. Setting of integration time (for fixed-wavelength measurement) 5. PC control Spectrophotometer can be controlled by an external PC. This function is also used when performing operation with the standard UVProbe software and the optional LabSolutions UV-Vis software provided. *A USB cable is required.

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SHIMADZU

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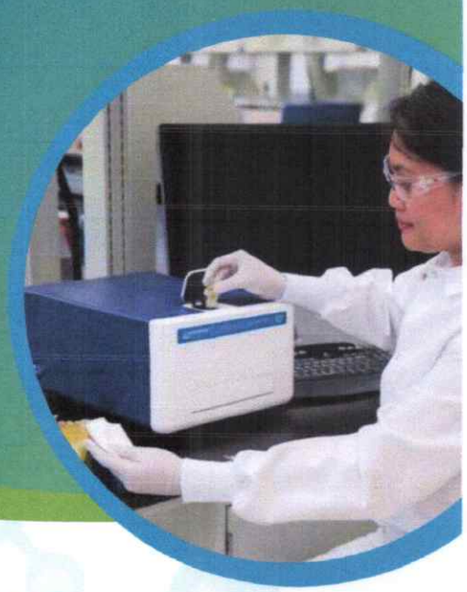
First Edition: April 2018

MOLECULAR DEVICES

DATA SHEET

SpectraMax ABS and ABS Plus Microplate Readers

Compact and tunable absorbance microplate readers



The SpectraMax® ABS Microplate Reader bridges the gap between the affordability of filter-based readers and the flexibility of monochromator-based systems.

The SpectraMax® ABS Plus Microplate Reader can run both cuvette-based and microplate reader applications on the same instrument. Read one sample or up to 384 in a single plate read using any standard cuvette, or 96- or 384-well microplate.

For more sample throughput, both readers can be easily integrated into full robotic systems and can easily adapt to fit sample processing requirements.

KEY FEATURES

- Compact design
- Minimum volume and maximum throughput with 96- or 384-well microplate compatibility
- Powerful data analysis with SoftMax Pro Software
- Optimal performance with validation tools

Tunable for filterless flexibility

The SpectraMax ABS readers use a grating monochromator to select the exact wavelength needed for every assay. The ABS reader ranges from 340–850 nm in wavelength and the ABS Plus reader ranges from 190–1000 nm, which is equivalent to having 510 or 810 free filters, respectively.

Exceptional performance

The advanced optical and electronic design of the readers give the same high performance with round-bottom, flat-bottom or half-area well plates. Select up to six wavelengths in a single read to maximize the number of data points collected in an experiment. Temperature control up to 45°C allows kinetic assays at ambient and physiological temperatures.

Patented PathCheck Sensor

The SpectraMax ABS Plus reader utilizes Molecular Devices PathCheck® Sensor—the only patented technology available that measures the depth (optical pathlength) of samples in a microplate. When used with SoftMax® Pro Data Acquisition and Analysis Software, it can automatically normalize the well absorbance to a cuvette equivalent pathlength of 1 cm. This is equivalent to having 96 or 384 cuvettes.

With the PathCheck Sensor, it is acceptable to have different volumes in the wells, so pipetting does not have to be accurate. The PathCheck Sensor will correct for the volume differences in all 96 or 384 wells and automatically report out 1 cm absorbance values.

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Extend the dynamic range to 6+ OD. If a well reads out of range (> 4 OD), decrease the volume in that well and re-read the plate using the PathCheck Sensor. A 100 µL sample (optical pathlength ~0.3 cm) that reads 2.8 OD will be corrected to ~9.2 OD.

Pipetting errors in 96- or 384-well microplates can be detected. The PathCheck Sensor measures differences in volume between the wells, so sources of error can be identified and eliminated from data, or absorbance values can be corrected.

Quickly and easily test multi-channel dispensers and pipettors. Measuring the depth of the liquid in the well, the PathCheck Sensor can determine the volume of liquid dispensed. Even 96- or 384-channel dispensers can be tested in a matter of minutes.

Wide range of applications

The SpectraMax ABS readers cover a wide range of applications:

- DNA quantitation (ABS Plus only)
- Microbial Growth/MIC
- IC_{50}/LD_{50}
- Endpoint ELISAs/EIA
- Cytoproliferation/Cytotoxicity
- Colorimetric Protein
- Kinetic ELISAs/Enzyme Assays
- Bacterial Identification
- Immunoassays
- Drug dissolution profiles
- Enzyme kinetics (e.g., Ki, Km, etc.)

Powerful data analysis

Industry leading SoftMax® Pro Data Acquisition and Analysis Software is included with both SpectraMax ABS readers and provides additional flexibility for the user. The software is designed to handle analysis requirements from simple endpoint assays to complex kinetic assays requiring custom calculations, meeting the needs of both basic and power users. The inclusive package of ready-to-run protocols, analysis algorithms, and 21 different curve fit options provides the full solution from data acquisition to analysis to publishing.

Validation made easy

The SpectraTest® ABS1 Absorbance Validation Plate tests optical performance using NIST-traceable standards. Testing can be done in the user's lab on their own schedule.

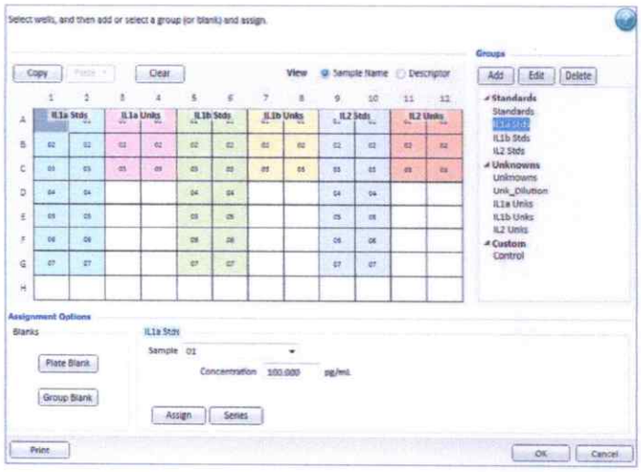


Figure 1. Flexible template assignment. Standards for multiple calibration curves and unknowns can be run on separate plates.

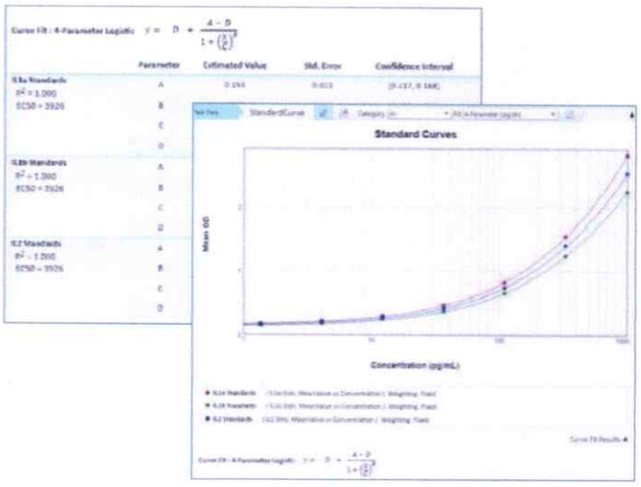


Figure 2. Multiple calibration curves. Multiple calibration curves can be plotted on one graph.

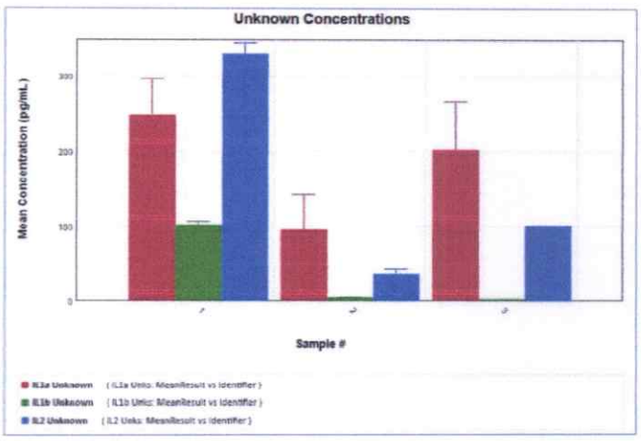


Figure 3. Combined data graphing. Results from unknowns run on different plates and different calibration curves can be plotted on one graph.

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Plate stacker and robot integration

The SpectraMax ABS readers can be integrated with Molecular Devices StakMax® Microplate Stacker within minutes and begin reading microplates with seven mouse clicks. For a higher degree of automation, the Automation Vendor Partners Program has streamlined the integration of our microplate reader systems with all leading partner robots. The "out-of-the-box" automation solution saves up-front integration time and resources.

Ordering information

Contact your Molecular Devices sales representative for configuration options.

General specifications	SpectraMax ABS/ABS PLUS Microplate Readers
Light source	Xenon flash lamp (5 Watts)
Average lamp lifetime	1 billion flashes
Illumination	Top down
Photodetectors	Silicon Photodiode
Plate shaking	Linear
Dimensions (in.) (H x W x D)	8 x 12 x 16.3
Dimensions (cm) (H x W x D)	20.3 x 30.5 x 41.4
Weight	25 lbs. (11.3 kg)
Power consumption	< 250 watts
Power source	100–240 Vac, 4 A 50/60 Hz

Technical specifications	SpectraMax ABS PLUS Microplate Reader with UV-Vis and Cuvette	SpectraMax ABS Microplate Reader
Photometric performance		
Wavelength range	190 nm – 1000 nm	340 nm – 850 nm
Wavelength selection	Monochromator, tunable in 1.0 nm increments	Monochromator, tunable in 1.0 nm increments
Wavelength bandwidth	2 nm	2 nm
Wavelength accuracy	±1.0 nm	±1.0 nm
Wavelength repeatability	±0.2 nm	±0.2 nm
Stray light	< 0.05% @ 230 nm	< 0.07% @ 340 nm
Photometric range	0.000 OD – 4.000 OD	0.000 OD – 4.000 OD
Photometric resolution	0.001 OD	0.001 OD
Photometric accuracy (microplate)	< ±0.006 OD ±1.0%, 0–3 OD	< ±0.006 OD ±1.0%, 0–3 OD
Photometric accuracy (cuvette)	< ±0.005 OD ±1.0%, 0–3 OD	N/A
Photometric precision	< ±0.003 OD ±1.0%, 0–3 OD	< ±0.003 OD ±1.0%, 0–3 OD
Read times (kinetic interval for 96-well plate)	5 seconds	8 seconds
Temperature regulation		
Temperature range	5°C above ambient to 45°C	5°C above ambient to 45°C
Temperature uniformity (microplate)	±0.75°C at 37°C	±0.75°C at 37°C

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Regional Offices

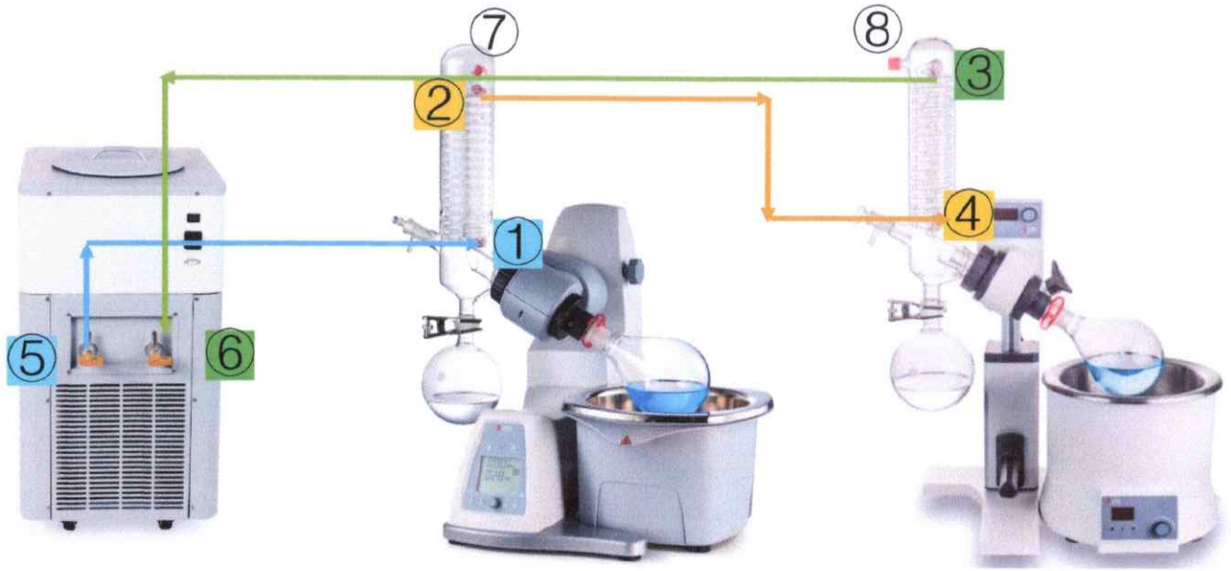
USA and Canada	+1.800.635.5577	China (Beijing)	+86.10.6410.8669	Japan	+81.120.993.656
United Kingdom	+44.118.944.8000	China (Shanghai)	+86.21.3372.1088	South Korea	+82.2.3471.9531
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5/20 2182G
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SCIP5-20 connecting with 2 units rotary evaporators

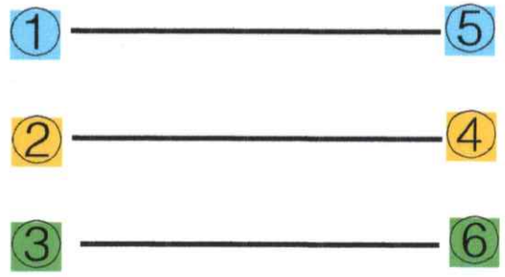


Low Temp. Circulator SCIP5-20

Rotary Evaporator SCI100-Pro

Rotary Evaporator SCI100-S

①	RE100-Pro cooling water inlet
②	RE100-Pro cooling water outlet
③	RE100-S cooling water outlet
④	RE100-S cooling water inlet
⑤	Circulator water outlet
⑥	Circulator water inlet
⑦	RE100-Pro vacuum port
⑧	RE100-S vacuum port





SHIMADZU
Excellence in Science

C054-E038E

Electronic Balances

UW/UX Series



UW/UX Series

Fast Response **Weighing takes just 0.7–1.2 seconds (S type)**

Excellent Durability **Long-life design employing a unique weighing mechanism**

Easy Connection **WindowsDirect function allows easy data communication to a computer**

Great for the Following Applications

High-precision balances with comprehensive functions handle diverse applications and increase productivity.

I want to upload measured data to a PC but installing the software is difficult.....

No software needed!

The WindowsDirect function transmits data to a PC with just a cable. Achieve remarkable improvements in work efficiency with no additional investment, except for a cable.
The standard auto print functions can be combined with the WindowsDirect function.

If you'd like to use "WindowsDirect" with "Windows 7","Windows VISTA" or a USB port, please contact our distributors.

I always need to weigh accurately but calibrating my balance is tedious.....

Easy calibration for every application

Calibrations are automatically performed upon detection of a temperature change (UW).
The balance automatically calibrates itself at the set times with its built-in calibration weights (UW).
Calibrations can also be made at any time with a simple key operation (UW).
Span calibration of the UX Series is easily performed by placing calibration weights on the balance according to the displayed instructions.

I need to manage weighing according to international standards.....

Use Shimadzu network-compatible software to ensure full compatibility with FDA 21 CFR Part 11.

Fully-automatic calibration ensures accurate weighing (UW). Just connect an optional printer to automatically output time-stamped hard copies of calibration reports.
LabSolutions Balance, Shimadzu's network-compatible database registration and management tool, enables full compatibility with FDA 21 CFR Part 11 (Electronic Records, Electronic Signatures).

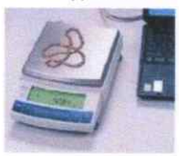
I work in a drafty, dark environment.....

Adjust the balance to match the environment

The display stability can be adjusted to the environment. Windbreak is provided as standard for models with a 1mg display, and is an option for other models. The auto zero function is useful in an environment where the TARE key must be pressed for every measurement. The back lighted display provides a clear display in dark environments.

WindowsDirect Function

Only a single cable is required to transfer balance data to MS Excel, or other Windows applications. No interface attachment or software installation is necessary. This unique function makes interfacing Shimadzu balances extremely convenient.

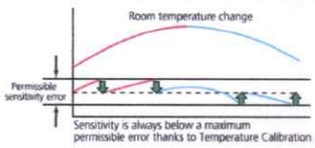


Automatic Touch-key Calibration (UW only)

Press just two keys to calibrate the balance whenever calibration is necessary.

PSC Fully-automatic Calibration (UW only)

Introduced in 1985, this user-friendly feature triggers calibration when a temperature change is detected, automatically maintaining the balance's sensitivity.



Clock-CAL Fully-automatic Calibration (UW only)

Automatic calibration at pre-set times. Up to three times a day can be programmed.



GLP/GMP/ISO Calibration Record

Connect the optional printer to automatically output calibration results.
Date and time are supplied by the balance's built-in clock.

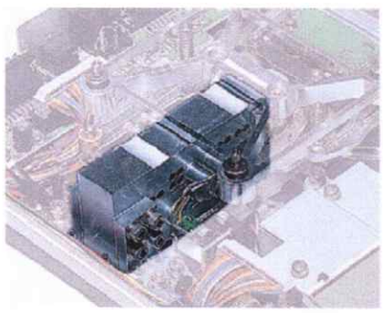


Backlight LCD

The backlight LCD display can be clearly read in the darkest of environments.

UniBloc Technology

UniBloc one-piece force cell technology replaces the core mechanism of a high-precision balance with a single aluminum alloy block. This provides a number of advantages, introducing fast response, high stability, and a long operational life.



What About Special Requirements?

I Have to Weigh a Lot of Valuable Samples Very Accurately

The improved UW/UX Series resolution achieves a minimum display accuracy of 1 mg for 1,020 g.

I Need to Weigh Large Objects

The large pan easily accommodates samples extending beyond the pan.
A built-in weigh-below hook allows weighing even unusally oversized items.

Can I Weigh Living Animals?

The standard animal-weighing mode allows accurate weighing of moving animals. It also gets the balance ready for the next animal without having to clean the pan each time.
An optional animal bucket is available.

I Have to Handle Chemicals

The stainless-steel pan can be easily removed for washing. The metal housing is resistant to organic solvents.
A clear, molded in-use cover is standard.

Resolution 1/1,020,000

High-resolution models with a resolution of 1,020 g/1 mg can accurately detect minute changes in a large mass.

Checkweighing Function

When upper and lower thresholds are set, the balance indicates if the sample weight is within the range (GO), over (HI) or under (LO).

Weigh-below hook

The standard weight-below hook enables weighing on a user's customized pan.

Specific Gravity Function

The standard specific gravity measurement calculation function uses the Archimedes principle. Attaching the optional specific gravity measurement kit allows the balance to be used as a density meter or a hydrometer.

Piece Counting

A piece counting function is standard in addition to various unit conversions.

I Need to Weigh a Lot of Objects.....

Offers Fast, Stable and Accurate Results

The ultra-fast response reduces operator stress and shortens work time.
UniBloc technology significantly enhances stability and durability, ensuring reliable weighing results over repeated operations.
If a weighing record is required, the standard auto print function can automatically output data each time a weighed sample is placed on the balance. In addition, all settings and functions can be controlled from a computer for compatibility with automated production-line operation.

UW/UX Series

Specifications

UW (with built-in calibration weights)

UW Series Model	UW220H	UW420H	UW620H	UW820H	UW1020H	UW2200H	UW4200H	UW6200H	UW420S	UW820S	UW4200S	UW8200S
Capacity (g)	220	420	620	820	1020	2200	4200	6200	420	820	4200	8200
Minimum Display (g)	0.001	0.001	0.001	0.001	0.001	0.01	0.01	0.01	0.01	0.01	0.1	0.1
Calibration Range with External Weights (g)	100–220	100–420	100–620	400–820	500–1020	1000–2200	1000–4200	1000–6200	100–420	100–820	1000–4200	1000–8200
Repeatability (σ) (g)	≤0.001					≤0.01			≤0.008		≤0.08	
Linearity (g)	±0.002			±0.003		±0.02			±0.01		±0.1	
Response Time (s)	1.5–2.5								0.7–1.2			
Ambient Temperature and Humidity	5–40°C 30–85% (No Condensation)											
Temperature Coefficient Sensitivity (ppm/°C) (10–30°C)	±3								±5			
Pan Size (mm) approx.	108 × 105					170 × 180			108 × 105		170 × 180	
Main Body Dimensions (mm) approx.	W190 × D317 × H78											
Weight (kg) approx.	3.4					4.6			3.4		4.6	
Display	LCD with backlight											
Power Requirements	DC, 10 to 15.5 V, 500 mA											
Data I/O	RS-232C											
Features	LabSolutions Balance WindowsDirect PSC Clock-CAL GLP/GMP/ISO conformance Analog display, % display, PCS, User unit, Animal weighing, Specific gravity measurement S/W, Checkweighing											

UX (Standard model)

UX Series Model	UX220H	UX420H	UX620H	UX820H	UX1020H	UX2200H	UX4200H	UX6200H	UX420S	UX820S	UX4200S	UX8200S
Capacity (g)	220	420	620	820	1020	2200	4200	6200	420	820	4200	8200
Minimum Display (g)	0.001	0.001	0.001	0.001	0.001	0.01	0.01	0.01	0.01	0.01	0.1	0.1
Calibration Range with External Weights (g)	100–220	100–420	100–620	400–820	500–1020	1000–2200	1000–4200	1000–6200	100–420	100–820	1000–4200	1000–8200
Repeatability (σ) (g)	≤0.001					≤0.01			≤0.008		≤0.08	
Linearity (g)	±0.002			±0.003		±0.02			±0.01		±0.1	
Response Time (s)	1.5–2.5								0.7–1.2			
Ambient Temperature and Humidity	5–40°C 30–85% (No Condensation)											
Temperature Coefficient Sensitivity (ppm/°C) (10–30°C)	±3								±5			
Pan Size (mm) approx.	108 × 105					170 × 180			108 × 105		170 × 180	
Main Body Dimensions (mm) approx.	W190 × D317 × H78											
Weight (kg) approx.	2.7					2.9			2.7		2.9	
Display	LCD with backlight											
Power Requirements	DC, 10 to 15.5 V, 500 mA											
Data I/O	RS-232C											
Features	LabSolutions Balance WindowsDirect GLP/GMP/ISO conformance Analog display, % display, PCS, User unit, Animal weighing, Specific gravity measurement S/W, Checkweighing											

Options	Peripherals	
UW/UX series	Electronic Printer EP-100 / EP-110	Stand for Remote Display (1m high)
	RS-232C Interface IFB-102A (needed only for multiple connection)	Comparator Lamps 100V (needs IFB-RY1)
	Small Size Windbreak (for models with capacity of 300g to 620g only) (Std. Acc. for models with readability of 0.001g)	Interface for comparator IFB-RY1 100V
	Glass Windbreak (for models with capacity of 220g to 820g only)	Foot Switch FSB-102PK (For printing)
	Large Size Windbreak (for all models)	Foot Switch FSB-102PK (For taring)
	Specific Gravity Measurement Kit SMK-101 (for Large size pan 170 × 180 mm)	RS-232C Cable, for IBM PC/AT Compatibles (25P-9P, Null modem, 1.5 m)
	Specific Gravity Measurement Kit SMK-102 (for Small size pan 108 × 105 mm)	RS-232C Cable, for multiple connections (25P-25P, Null modem, 1.5 m)
	Protective in-use cover for key panel and display (5 pcs)	Application Keyboard AKB-301
	Small Animal Bucket set (For large pan models only)	Remote Display Unit RDB-201 with operation keys
	Angle Adjuster and Wall Hook for Remote Display	Remote Display Unit RDB-202

SHIMADZU

Shimadzu Corporation

www.shimadzu.com/an/

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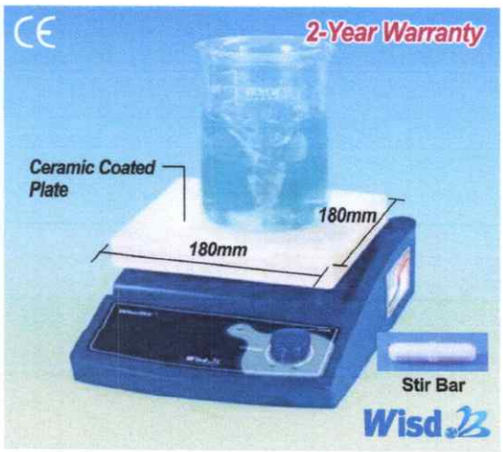
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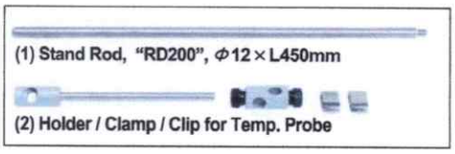
Articles	Cat. No	Description

DAIHAN-brand® Premium Magnetic Stirrer, "MS-A", 180×180mm Ceramic-Coated Plate, with Certi. & Traceability with PWM Control System, Precise Speed Control, Ceramic Coated Al-Plate, up to 1,500 rpm, Max. 20 Lit.

아날로그 자력교반기, PWM 컨트롤 시스템, 세라믹 코팅 플레이트



Premium Magnetic Stirrer with Stir Bar "MS-20A"



- **with Certificate & Traceability : Controlled by Serial Number, Certificate, Delivery-information, and Traceable Data Base System**
- **Ceramic Coated Plate-Chemical/Acid Resistance**
- **CE Certified**
- **PL (Product Liability) Insurance**

* Ideal for Heat-sensitive Procedures such as Microbiology and Biochemistry Applications
* Mixing for Chemical, Laboratory, Pharmaceutical Materials
* Superior Linearity for the Control of Stirring Speed by Specially Designed Control Circuitry (PWM Control)
* Smart Design of Body : Made by Aluminum Die-Casting
* Powerful Motor for Stirring Quantities of up to 20Lit. (Water)
* Ceramic Coated Plate-Chemical/Acid Resistance
* Free Voltage (AC 80V ~ 240V, 50 / 60Hz) by SMPS
* Included 1ea 3cm Magnetic Stirrer Bar

Specification

Model		180×180mm Plate-type "MS-20A"
Plate Size (mm)		180×180
Stirring Capacity (H_2O)		Max. 20Lit.
Speed Range		Max. 1,500 rpm
Motor Rating		9 / 4 W
Controller		Superior Linearity PWM Controller
Stirring Bar		3cm bar-included, Usable : up to 5cm bar
Permissible Ambient Temperature		Ambient +5 ~ 50℃
Permissible Relative Moisture		85%
Material	Plate	Ceramic Coated Aluminum (Chemical / Acid Resistance)
	Body	Powder Coated Aluminum
Dimension (w×d×h) (mm)		206×307×99
Weight (kg)		3.1 kg
Others		CE Certified PL (Product Liability) Insurance
Packing Size (w×d×h) (mm) & Gross Weight		400×280×200 5 kg
Power Consumption		10 W
Power Supply & Cord/Plug*		Free Voltage (1 Phase, AC 80 ~ 240 V, 50 / 60 Hz)

* Other Specifications are available upon Customer's Request.

Specialized Features

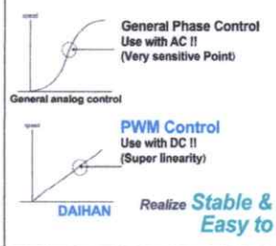
Long Life Power & Mechanism!

SMPS (Switching Mode Power Supply)
Use with Safety & Regular DC !!



SSR (Solid State Relay)-Non Contact
No Electronic Harm!
Realize ***Long Life Cycle!***

High Stability in Very Low & High-speed



Realize ***Stable & Easy to Use!***

Ordering Information

		Ea. Price, ₩
DH.WMS03122	Magnetic Stirrer, 180×180mm, **230V, "MS-20A"**, with stir bar	189,120
DH.WMS02122	, **120V, "MS-20A"**, with stir bar	189,120

Optional Accessories

		Ea. Price, ₩
DH.WHP503036	(1) Stand Rod, **"RD200"**, Stainless steel, screw, Φ12×L450mm, for **"HP-20A/20D", "MSH-20A/20D"**, and **"SMHS"**	17,820
DH.WHP503012	(2) Holder/Clamp/Clip, **"CL220"**, for Temp. Probe	23,540

Centrifuge 5430/5430 R

Crossover microcentrifuges with small footprint and great versatility. Refrigerated for temperature sensitive samples or non-refrigerated.

Centrifuge 5430 and 5430 R combine the dimensions of a microcentrifuge with the application range of a multipurpose centrifuge in one instrument: 12 rotor options including rotor for 15 and 50 mL conical tubes, 30-place high speed rotor and swing-bucket rotor for MTPs or DWPs.





Product features

> Capacity: 48 x 1.5/2.0 mL, 6 x 50 mL, 2 x MTP
> Speed up to 30,130 x *g* (17,500 rpm)
> Soft-touch closure for ergonomic operation
> Menu-driven, multi lingual operation menu (English, German, French, Spanish) with large backlit display
> Saves up to 50 user-defined programs – 5 program keys for easy access to routine programs

Additional features of refrigerated Model 5430 R

> Temperature control range from –11 °C to 40 °C
> FastTemp pro® control program for fast pre-cooling and maximum temperature accuracy inside the rotor
> ECO shut-off can be programmed to engage after 1, 2, 4, or 8 hours of nonuse for up to 84 % energy savings and extended compressor life

Ordering information

Description	Order no.
Keypad	
Centrifuge 5430, non-refrigerated, without rotor, keypad, 230 V/50 – 60 Hz	*
Centrifuge 5430, non-refrigerated, with Rotor FA-45-30-11 incl. rotor lid, keypad, 230 V/50 – 60 Hz	*
Centrifuge 5430 R, refrigerated, without rotor, keypad, 230 V/50 – 60 Hz	*
Centrifuge 5430 R, refrigerated, with Rotor FA-45-30-11 and rotor lid, keypad, 230 V/50 – 60 Hz	*
Rotary knobs	
Centrifuge 5430, non-refrigerated, without rotor, rotary knobs, 230 V/50 – 60 Hz	*
Centrifuge 5430, non-refrigerated, with Rotor FA-45-30-11 incl. rotor lid, rotary knobs, 230 V/50 – 60 Hz	*
Centrifuge 5430 R, refrigerated, without rotor, rotary knobs, 230 V/50 – 60 Hz	*
Centrifuge 5430 R, refrigerated, with Rotor FA-45-30-11 and rotor lid, rotary knobs, 230 V/50 – 60 Hz	*

* For ordering information, please check our local website or contact your local sales representative.
www.eppendorf.com/centrifuge-5430

Rotors for Centrifuge 5430/5430 R



Rotor FA-45-48-11
> Max. speed: 18,210 x *g* (12,700 rpm)
> Max. capacity: 48 x 1.5/2.0 mL tubes
> Aerosol-tight for safe centrifugation of hazardous samples
> Eppendorf QuickLock® lid takes repetitive stress off your wrist



Rotor F-45-48-11
> Max. speed: 18,210 x *g* (12,700 rpm)
> Max. capacity: 48 x 1.5/2.0 mL tubes



Rotor S-24-11-AT
> Max. speed: 16,049 x *g* (12,700 rpm)
> Swing-bucket rotor for 24 x 1.5/2.0 mL tubes
> Aerosol-tight with Eppendorf QuickLock® lid
> Perfect for phase-separation applications such as phenol-chloroform extraction



Rotor FA-45-16-17
> Max. speed: 21,191 x *g* (14,200 rpm)
> Max. capacity: 16 x Eppendorf Tubes® 5.0 mL
> Aerosol-tight for safe centrifugation of hazardous samples
> Eppendorf QuickLock® lid takes repetitive stress off your wrist
> Eppendorf Tubes® 5.0 mL with screw caps only can use up half (8 x) of the places in the rotor



Rotor FA-45-30-11
> Max.speed: 20,817 x *g* (14,000 rpm)
> Max. capacity: 30 x 1.5/2.0 mL tubes
> Aerosol-tight for safe centrifugation of hazardous samples
> Eppendorf QuickLock® lid takes repetitive stress off your wrist
> PTFE coating for increased chemical resistance



Rotor FA-45-24-11-Kit
> Max. speed: 19,090 x *g* (13,200 rpm)
> Rotor with higher edge for safe centrifugation of up to 24 spin columns
> Aerosol-tight for safe centrifugation of hazardous samples
> Eppendorf QuickLock® lid takes repetitive stress off your wrist



Rotor F-45-30-11
> Max. speed: 20,817 x *g* (14,000 rpm)
> Max. capacity: 30 x 1.5/2.0 mL tubes
> PTFE coating for increased chemical resistance



Rotor FA-45-24-11-HS
> Max.speed: 30,130 x *g* (17,500 rpm)
> Max. capacity: 24 x 1.5/2.0 mL tubes
> Extremely fast separation results
> Aerosol-tight for safe centrifugation of hazardous samples
> Eppendorf QuickLock® lid takes repetitive stress off your wrist
> PTFE coating for increased chemical resistance



Rotor F-35-6-30
> Max. speed: 7,745 x *g* (7,830 rpm)
> Rotor for 6 x 15/50 mL conical or 12 x BD Vacutainer® tubes



Rotor A-2-MTP
> Max. speed: 2,204 x *g* (4,680 rpm)
> 2-place swing-bucket rotor for MTP, PCR and Deepwell Plates (max. height = 29 mm)
> Windshield design for quiet operation



Rotor F-45-64-5-PCR
> Max. speed: 13,543 x *g* (11,800 rpm)
> PCR strip rotor for up to 64 x 0.2 mL PCR tubes



Rotor F-45-18-17-Cryo
> Max. speed: 8,324 x *g* (8,900 rpm)
> 18-place rotor for Cryovial® tubes

Rotors for Centrifuge 5430/5430 R

Ordering information

Description	Order no.
Rotor FA-45-16-17, aerosol-tight, incl. rotor lid, for 16 x Eppendorf Tubes® 5.0 mL	5427 750 002
Rotor FA-45-48-11, aerosol-tight, incl. rotor lid	5427 754 008
Rotor F-45-48-11, incl. rotor lid	5427 755 004
Rotor S-24-11-AT, aerosol-tight, incl. rotor lid	5427 757 007
Rotor FA-45-30-11, aerosol-tight, incl. rotor lid	5427 753 001
Rotor FA-45-24-11-Kit, aerosol-tight, incl. rotor lid	5427 752 005
Rotor F-45-30-11, incl. rotor lid	5427 712 003
Rotor FA-45-24-11-HS, aerosol-tight, incl. rotor lid	5427 710 000
Rotor F-35-6-30, incl. rotor lid, with 6 adapters each for 15 mL/50 mL conical tubes	5427 716 009
Rotor A-2-MTP, incl. wind shield and rotor lid, with 2 buckets	5427 700 005
Rotor F-45-64-5-PCR, incl. rotor lid and adapters	5427 714 006
Rotor F-45-18-17-Cryo, incl. rotor lid and Cryovial® adapters	5427 705 007
Adapter, for 1 tube 0.5 mL or BD microtainer®, max. Ø 8 mm, for all 1.5/2.0 mL rotors, set of 6 pcs.	5425 716 001
Adapter, for 1 PCR tube 0.2 mL, max. Ø 6 mm, for all 1.5/2.0 mL rotors, set of 6 pcs.	5425 715 005
Adapter, for 1 cryo containers (max. Ø 13 mm) or lidded centrifugation tubes (Ø 12.2 mm), max. length 50 mm, for Rotor F-45-18-17-Cryo, set of 6 pcs.	5702 752 002
Adapter, for 1 vessel 1.5 – 2.0 mL, for all 5.0 mL rotors, set of 4 pcs.	5820 768 002
Adapter, for 1 cryo tube, for all 5.0 mL rotors, set of 4 pcs.	5820 769 009
Adapter, for 1 HPLC tube, for all 5.0 mL rotors, set of 4 pcs.	5820 770 007
Adapter, for 1 tube 0.4 mL, max. Ø 6 mm, for all 1.5/2.0 mL rotors, set of 6 pcs.	5425 717 008
Adapter, for 1 HPLC vessel 1.5 mL, for Rotor F-45-18-17-Cryo	5427 708 006

Adapter options 6 x 15/50 mL conical tube rotor

Adapter	Rotor capacity[3]	Max. tube dimensions Ø x L	Max. RCF	For bore hole	Order no.[4]
13 mm round	12	13 x 65 – 89 mm	6,443 x g	small	5427 740 007
			6,100 x g	large	5427 742 000
13 mm round	12	13 x 90 – 110 mm	7,471 x g	small	5427 741 003
			7,129 x g	large	5427 743 006
16 mm round	12	16 x 75 – 105 mm	7,005 x g	small	5427 732 004
		16 x 90 – 105 mm	7,005 x g	large	5427 734 007
17 mm round	12	17 x 90 – 125 mm	7,745 x g	small	5427 735 003
			7,403 x g	large	5427 738 002
Eppendorf Tubes® 5.0 mL		17.5 x 80 mm	6,237 x g	small	5427 746 005
		17.5 x 80 mm	7,129 x g	large	5427 747 001
15 mL conical	6	17.5 x 100 – 125 mm	7,540 x g	small	5427 726 004
30 mL round	6	26 x 92 – 112 mm	7,087 x g	large	5427 736 000
50 mL round	6	29 x 95 – 125 mm	7,581 x g	large	5427 737 006
50 mL conical	6 [1) 2)]	29.5 x 100 – 125 mm	7,197 x g	large	5427 727 000
Centriprep®	6	29.5 x 100 – 125 mm	7,567 x g	large	5427 723 005

[1)] Three set of 2 included within 6 x 15/50 mL rotor package. [2)] Adapter accommodates skirted 50 mL conical tubes. [3)] Please perform a manual test to determine the actual loading capacity. [4)] Set of 2

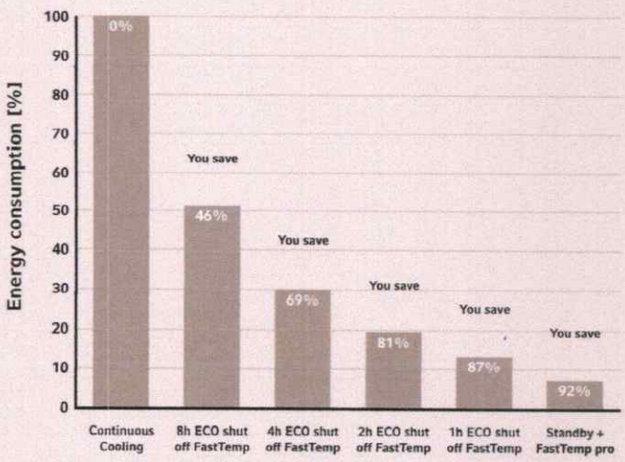
Centrifuge 5910 R and Centrifuge 5920 R

Advanced temperature management

Eppendorf developed the Dynamic Compressor Control (DCC). Unlike other compressors which switch back and forth between 0% and 100% performance output, DCC allows for a constant and smooth temperature regulation. This results in a precise cooling performance and the set temperature is kept stable at all time.

In addition to the well-known FastTemp function for super fast pre-cooling, both instruments include the new FastTemp pro® software option. This feature allows for automated pre-cooling based on pre-programmable time and date. Simply turn the centrifuge into standby mode when you leave your lab and let FastTemp pro take care of pre-cooling in the morning so you don't have to. This saves you more than 90% of energy over night and programming is super easy.

Centrifuge 5920 R: Energy saving overnight with ECO shut-off and FastTemp pro®





Innovative safety features

Centrifuge 5910 R and Centrifuge 5920 R have an automatic rotor recognition. This feature detects a newly inserted rotor and automatically limits the centrifuge speed to the rotor's maximum permissible value.

When handling substances that pose a health hazard it is highly recommended to use aerosol-tight fixed-angle rotors with Eppendorf QuickLock® lids or swing-bucket rotors in combination with Eppendorf QuickLock caps.





> Automatic rotor recognition and imbalance detection for maximum operational safety
> Sturdy motor driven lid locks to ensure secure operation
> Not one but two gas springs to support the lid
> Aerosol-tight rotors and buckets to avoid contamination





Centrifuge 5910 R and Centrifuge 5920 R

Excellent user convenience

For ergonomic lid operation Eppendorf designed the soft-touch lid closure system. There is no need to exert unnecessary force when closing the centrifuge lid. In combination with a low opening height you can close the centrifuge with ease.

Centrifuge 5910 R and Centrifuge 5920 R feature an advanced operation system that comes with an easy to use interface to make the operation intuitive and more enjoyable. The multilingual operation menu includes 5 program keys for easy access to your routine programs.





> Soft-touch lid for stress-free instrument closing
> Low access height for easy loading and unloading
> Low noise level to improve your work environment
> Large display and intuitive operating system
> 5 program keys for easy access to routine programs

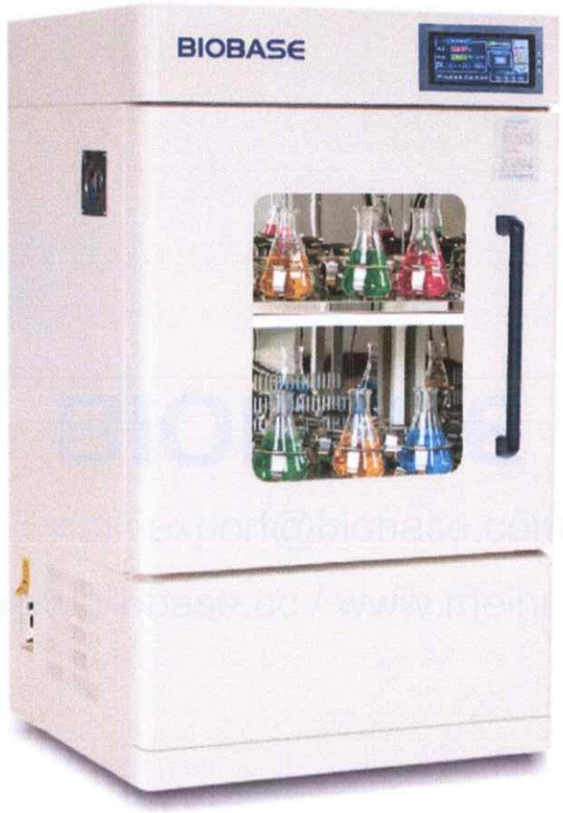






Feature	Centrifuge 5910 R	Centrifuge 5920 R
1,000 mL Eppendorf bottles (see pages 14–15 for more information)	4	–
1,000 mL standard bottles	–	4
750 mL bottles (see pages 14–15 for more information)	4	4
180 – 250 mL bottles (flat)	4	8
50 mL conical tubes	36	52
25 mL conical tubes	36	52
5 mL and 15 mL conical tubes	68	108
1.5/2.0 mL Eppendorf Tubes®	200	368
BD Vacutainer® CPT™ (16 mm × 125 mm/8 mL)	28	80
16 mm round-bottom and blood collection tubes	104	148
13 mm round-bottom and blood collection tubes	120	196
PCR plates/MTP	20/4	28/8
Universal adapters for conical tubes, plates and bottles	■	■
Dual use of plates and tubes in one rotor bucket	■	■
Plate rotor options for MTP, PCR or DWP	■	■
Fixed-angle rotors for high-speed applications	■	■

Vertical Type Shaking Incubator (Single Door & Double Layer)



Features:

* Audio and visual alarm.
* Power off recovery function.
* High efficiency servo motor, accurate control speed.
* Super low speed start, start speed adjustable, overspeed automatic protection.
* Accessories:Monitor timer (standard accessories); Open the door automatically stop, UV disinfection function, Humidification shaker, high temperature shaker(optional accessories).

Technical Parameters:

Model	BJPX-1102D	BJPX-2102D
Controller	PID	
Display	LCD	
Circulation Mode	Forced convection	
Shaking Mode	Rotary shaking	
Drive Mode	Multidimensional drive	
Rotating Speed	20-350rpm(Static culture, reverse rotation or alternate operation)	
Rotating Accuracy	±1rpm	
Vibration Range	Φ26mm(Can customize φ35 (20-250rpm/min); φ50(20-200rpm/min))	
Temp. Range	RT+5~65℃	4~ 60℃
Temp. Accuracy	±0.1℃	
Temp. Uniformity	±0.5℃(At 37℃)	
Timing Range	0~9999h/min(Can run continuously from time to time)	
Shaking Plate Size	500*360mm	
Shaking Plate Quantity	2pcs	
Max. Capacity	100ml*60or250ml*38or500ml*22or1000ml*16(Flask clamp) 100ml*60or250ml*36or500ml*24or1000ml*16(Haftplatte)	
Standard Capacity	250ml*24	
Internal Material	Stainless steel	
External Material	Stainless steel	
Refrigerant	/	CFC-Free,R134A
Cooling Consumption	/	250W
Heating Consumption	500W	
Safety Function	Audio and visual alarm for over-temperature; Alarm for over-speeds; Independent over-temperature protector; Shutdown protection for door opening; (Optional accessories) Leakage or over-current protection; Ultraviolet disinfection function (Optional accessories); Calibration of monitoring temperature and ambient temperature.	
Consumption	600W	850W
Power Supply	AC220V±10%, 50/60Hz(Standard); AC110±10%, 60Hz(Optional)	
Internal Size(W*D*H)(mm)	600*410*630	
External Size(W*D*H)(mm)	700*580*1280	
Package Size(W*D*H)(mm)	825*720*1490	
Gross Weight(kg)	220	

BIOBASE®

Oven

Constant-Temperature Drying Oven

Features:

* External material: cold-rolled steel with anti-bacteria powder coating.
* Stainless steel inner chamber, round angle structure, adjustable shelves.
* PID control with LED display, soft touch operational key, timing function and over-temperature protection.
* Door with double-layer tempered glass observation window, adjustable air-tight buckle lock, and silicon rubber seal, which provide good seal ability and efficiently prevent heat loss.



Technical Parameters:

Model	BOV-T30C	BOV-T70C	BOV-T140C	BOV-T270C
Capacity	43L	72L	137L	270L
Temp. Range	50~200℃			
Temp. Fluctuation	±1℃			
Ambient Temp.	5~40℃			
Timing Range	1~9999min			
Shelves No.	2 pcs		3 pcs	
Consumption	850W	1000W	2000W	3000W
Power Supply	AC110V/220V±10%, 50/60Hz			
Internal Size(W*D*H) mm	350*350*350	400*400*450	500*500*550	600*600*750
External Size(W*D*H) mm	630*540*550	720*570*750	820*670*850	920*770*1050
Package Size(W*D*H) mm	780*750*700	870*700*800	960*780*950	1060*880*1150
Gross Weight(kg)	60	70	105	145

Ducted PP Fume Hood





LED Display



Waterproof Socket



PP Sink
resistant to acid, alkali and anti-corrosion



Water&Gas Remote Control

ADVANTAGE

1. It is safer to use anti-corrosive water tap.
2. Microprocessor control system, LED display.
3. With memory function in case of power-failure.
4. Made of porcelain white PP, resistant to acid, alkali and anti-corrosion.
5. Front window which is made of thick transparent toughened glass maxi fume hood, providing a bright and open working environment.

TECHNICAL PARAMETERS

Model		FH1000(P)	FH1200(P)	FH15
External Size (W*D*H)		1047*800*2450mm	1247*800*2450mm	1547
Internal Size (W*D*H)		787*560*700mm	987*560*700mm	1287
Work Surface Height		820mm		
Max Opening		740mm		
Air Velocity		0.3~0.8m/s		
Noise		≤68dB		
Illuminating Lamp		LED lamp		
		12W*1	30W*1	30W*
Blower		Built-in PP centrifugal blower (2 blowers for FH1800		
Front Window		Resistant to acid and alkali, Manual, 5mm toughene		
Power Supply		AC220V±10%, 50/60Hz; 110V±10%, 60Hz		
Consumption		330W	360W	360W
Material	Main Body	Made of Porcelain white PP, thickness 8mm, resista		
	Work Table	Chemical resistant phenolic resin		
Standard Accessory		Illuminating lamp, Water tap, Gas tap, Water Sink, B PVC duct (2 Pcs of 4 meters PVC duct for FH1800(PP centrifugal blower pipe strap*2 (4 Pcs for FH1800(P) only)		
Optional Accessory		PP work table, epoxy resin board or ceramic board, . Outer PVC centrifugal blower (only need to select wi		
Gross Weight		225kg	253kg	294k
Package Size (W*D*H)	Main Body	1188*938*1612mm	1388*938*1612mm	1688
	Base Cabinet	1188*888*1000mm	1388*888*1000mm	1688

Innovative Technology to Improve the Quality of Human Life

Features of New Steam Sterilizers, Digital Fuzzy Controlled Autoclaves

DAIHAN New Steam Sterilizers

DAIHAN's new Autoclaves are upgraded with a simpler and more ergonomic design. Unique and innovative 'Electronic Safety Door Locking System' and 'Steam Condensing Mechanism' maximize the user safety, which is the most critical feature of autoclave product. New Lever-type Door Handle makes anybody open and close the door very easily and safely. Both of Solid and Liquid cycles are offered up to the samples to be sterilized, which ensures optimum sterilizing performance.



New Steam Sterilizers,
Digital Fuzzy Controlled Autoclaves

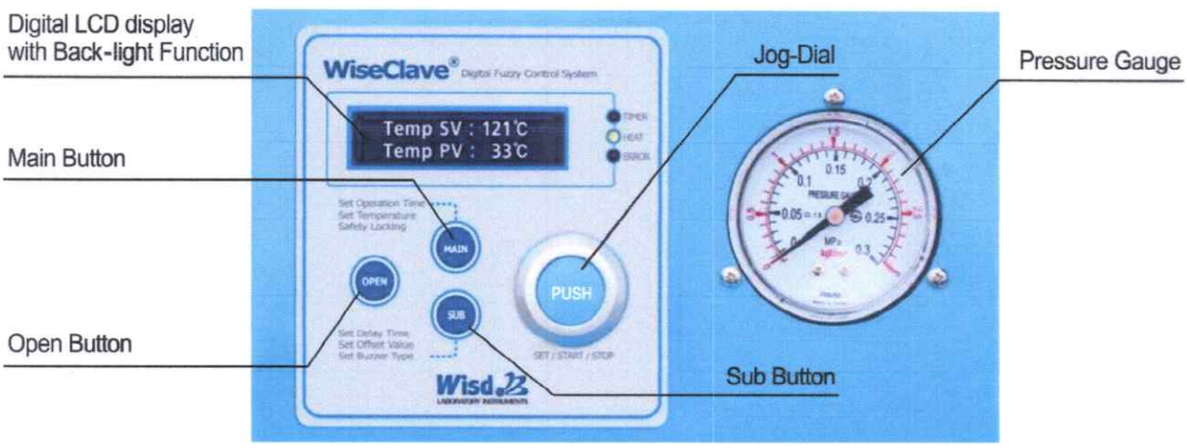
>> Safety Upgrade

- Special **'Steam Condensing Mechanism'** protects the user from hot steam after sterilizing cycle, by condensing the hot steam into small amount of water : patent registered.
- **'Safety Door Locking System'** ensures the door firmly locked by electronic-controlled switch and solenoid latch while sterilizing cycle is ongoing. The door can be opened only when there is no hot-steam and high-pressure inside the pressure vessel: patent registered.
- **CE-MDD (0470)** certified as a medical device, by which DAIHAN's autoclaves can be supplied to the hospitals in Europe.
- Stainless Steel Cover is implemented to prevent the user from burn by high temperature in the pressure vessel.
- Very high quality **Over Pressure Valve** is used to automatically release abnormal high pressure inside the vessel.

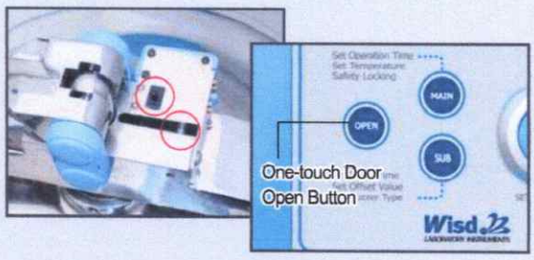
>> Function/Convenience Upgrade

- **One-Touch Lever-type Door Handle** is invented and implemented for the user to open & close the door very easily and safely : patent registered.
- Two different operation mode of **Solid and Liquid Cycle** are offered, which can be chosen up to the kind of sample to be sterilized.
- New Jog-Dial Controller with simples user interface: patent registered
- Digital Timer functions for Delay time and Operation time are available.
- Innovative Easy-to-Use Digital Fuzzy Controller.
- Digital LCD with Back-Light

>> New Jog-Dial Controller

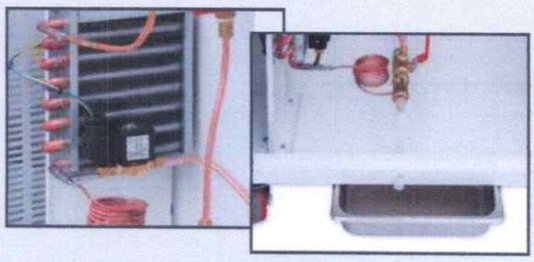


- Simplest and easiest User Interface by ergonomic design
- New design of Jog-Dial is more sturdy and convenient
- High quality Digital LCD with Back-Light
- One-touch 'Open button' to open the door only when sterilizing cycle is not on going



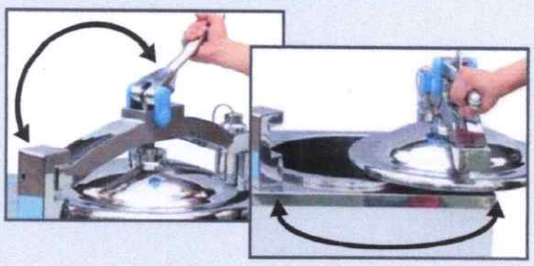
Safety Electronic Door Locking System

Sterilizing cycle can be stated only when the door is completely closed. Eclectro-mechanical sensor, solenoid latch, and one-touch Open button work together to implement the perfect safety door, which can be opened only when the temperature and pressure is low enough inside the pressure vessel.



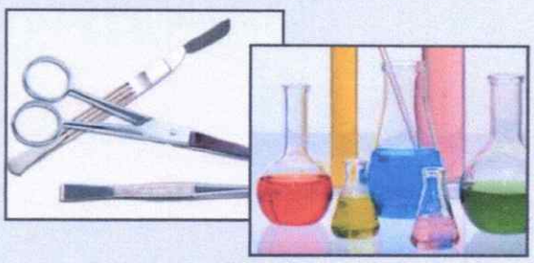
Special Steam Condensing Mechanism

Hot steam generated by sterilizing cycle is not released out directly.
It is condensed into a small amount of water just after the sterilizing cycle by special condensing mechanism, which protects the user from dangerous high temperature steam.



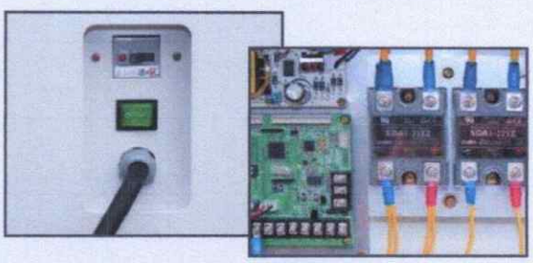
Lever-type Sliding Door

This real innovation of door closing mechanism makes any user easily close the door with consistent degree of closing by a single hand, regardless of the level of strength in his/her arms. Opening is also very easy and can be done with single hand as well.



Liquid & Solid Cycle Mode

Operation mode is optional between Solid Cycle mode and Liquid Cycle mode. In Solid Cycle mode, heating is immediately off and steam is released quickly just after sterilizing period. But in Liquid mode, temperature is decreased very slowly and steam is released out little by little for much longer time.



Using International Standard Components

Maintenance and Service is very convenient because most of the components and parts are internationally standard ones and available in the market.

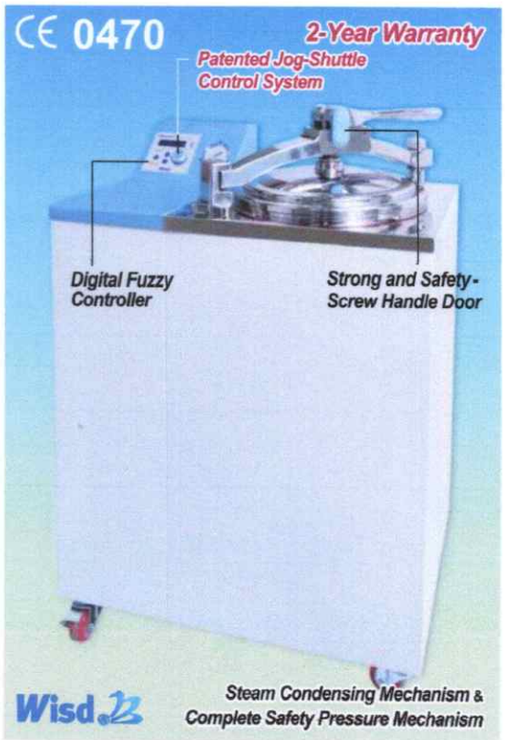


CE-MDD(CE 0470-Medical Device Directive) Certified

DAIHAN's new autoclaves are designed and produced strictly following European standard for electric and mechanical safety. So, it can be supplied to, and used safely in, world-wide hospitals and institutes requiring CE-MDD Certification.

DAIHAN® Steam Sterilizers, Digital Fuzzy-controlled Autoclaves, "WAC", 47-/60-/80-/100-Lit., up to 132℃, Max.2 kgf/cm²

Electronic Door Lock, Steam Condensing Mechanism, Lever-type Sliding Door, Solid-/Liquid-Modes, CE-MDD Certified



Steam Sterilizers, Digital Fuzzy-Controlled Autoclave, With 2× Wire Baskets

- **with Certificate & Traceability : Controlled by Serial Number, Certificate, Delivery-information, and Traceable Data Base System**
- **Patented Special Steam Condensing Mechanism**
- **Patented Jog-Shuttle Control System**
- **"CE-MDD (CE 0470 - Medical Device Directive) Certified"** (WAC-100 excluded)
- **PL (Product Liability) Insurance**

* Ideal for Biotechnology, Clinical, Environmental, Medical and Food Industry.
* Innovative Easy-to-Use Digital Fuzzy Controller.
* Top-loading type Sterilizer for Easy Loading of Large Working Volume.
* Casters equipped for Excellent Mobility. * Digital LCD Display with Back-Light Function.
* Digital Timer Function - 'Delay Time' and 'Operation Time' can be Controlled.
* **Solid / Liquid Mode** can be chosen by a customer considering a Sample Phase.
* Strong and Safe Screw-type Handle Door. * Complete Safety Pressure Mechanism.
* Automatic Over Pressure Release Valve ensure User Safety.
* **Electronic Door Lock Sensing** System for User Safety.
* Heat-Protective Lid Cover for User Safety.

Specification

Capacity & Models		47 Lit. "WAC-47"	60 Lit. "WAC-60"	80 Lit. "WAC-80"	100 Lit. "WAC-100"
Temp. Range & Accuracy		Ambient 5℃ to 132℃, ±0.5℃ at 121℃			
Pressure & Gauge		1.2 kgf/cm² at 121℃ / 2.0 kgf/cm² at132℃, 0 ~ 3 kgf/cm²			
Sterilization mode		Solid- or Liquid-mode			
Heating Power/Consumption		2 kW	3 kW	2 kW ×2	2 kW ×2
Timer & Alarm		99 hr 59 min 59 sec, Error status & Timer-end			
Display		Digital LCD Display with Back-Light Function			
Controller		Digital Fuzzy-Control System (Fully Automatic)			
Door & Closing System		Lever-type Sliding Door / Safety Electronic Door Closing System			
Safety Circuit		Over-Temp./-Pressure/-Current protection, Sensor Error Detection			
Material		Stainless Steel (#304)-Pressure Vessel, Powder Coated Steel-External			
Basket		2ea of Stainless Steel Wire Baskets are Included			
Basket Dimension (mm)		Φ270×h240	Φ320×h240	Φ370×h240	Φ420×h240
Dimension (w×d×h)	Inner (mm)	Φ300×h670	Φ350×h650	Φ400×h650	Φ450×h630
	Outer (mm)	680×470×880	680×470×900	780×580×900	780×580×900
Packing Size (mm)		800×700×h1,280 mm		940×800×h1,350 mm	
Gross Weight (kg)		116 kg	120 kg	173 kg	173 kg
Power Supply*		1Phase - AC 120V, 60Hz or - AC 230V, 50/60Hz		1 Phase - AC 120V, 60Hz or - AC 230V, 50/60Hz 3 Phase with Terminal Block	

* Other Specifications are available upon Customer's Request.

Ordering Information

* 2ea of Wire Baskets are included in the All Autoclaves

Autoclaves, 1 Phase, Digital Fuzzy Controlled

DH.WAC05047 Autoclave, **230V, 47 Lit., "WAC-47"**, with Cord / Plug
DH.WAC05060 , **60 Lit., "WAC-60"**, with Cord / Plug
DH.WAC05080 , **80 Lit., "WAC-80"**, with Cable
DH.WAC05100 , **100 Lit., "WAC-100"**, with Cable

DH.WAC06047 Autoclave, **120V, 47 Lit., "WAC-47"**, with Cable
DH.WAC06060 , **60 Lit., "WAC-60"**, with Cable
DH.WAC06080 , **80 Lit., "WAC-80"**, with Cable
DH.WAC06100 , **100 Lit., "WAC-100"**, with Cable

Autoclaves, 3 Phase with Terminal Block for Custom-made Order

DH.WAC07080 Autoclave, **80 Lit., "WAC-80"**, with Terminal Block
DH.WAC07100 , **100 Lit., "WAC-100"**, with Terminal Block

<Continued on next page>

<Continued from... Steam Sterilizers, Digital Fuzzy-controlled Autoclaves, "WAC">

Ordering Information

(1) Spare Wire Baskets, Stainless Steel

DH.Ba7011	Wire Basket, Φ270×h240mm, **"Ba7011"**, for **47 Lit. WAC-47**
DH.Ba7012	, Φ320×h240mm, **"Ba7012"**, for **60 Lit. WAC-60**
DH.Ba7013	, Φ370×h240mm, **"Ba7013"**, for **80 Lit. WAC-80**
DH.Ba7014	, Φ420×h240mm, **"Ba7014"**, for **100 Lit. WAC-100**

(2) Optional Perforated Bottom Baskets, Stainless Steel

DH.WAC20047	Perforated Basket, Φ270×h250mm, **"WAC20047"**, for **47 Lit. WAC-47**
DH.WAC20060	, Φ320×h250mm, **"WAC20060"**, for **60 Lit. WAC-60**
DH.WAC20080	, Φ370×h250mm, **"WAC20080"**, for **80 Lit. WAC-80**
DH.WAC20100	, Φ420×h250mm, **"WAC20100"**, for **100 Lit. WAC-100**

Validation (IQ, OQ) Service

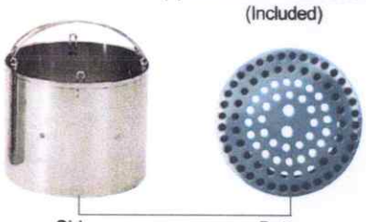
DH.WAC30047	Validation Service(IQ, OQ), **"ACV47"**, for **WAC-47**
DH.WAC30060	. **"ACV60"**, for **WAC-60**
DH.WAC30080	, **"ACV80"**, for **WAC-80**
DH.WAC30100	, **"ACV100"**, for **WAC-100**



Steam Sterilizers, Digital Fuzzy-Controlled Autoclave



(1) Wire Basket Stainless Steel
(Included)



(2) Perforated Bottom Basket, Stainless Steel
(Optional)